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Protocol choices contribute to variability in reported biochar impactson soil organic carbon: a reproducible open-data study withBiocharDS

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Abstract:	Global syntheses of biochar impacts on soil organic carbon (SOC) are increasingly used to justify climate mitigation and soil management, yet reported effect sizes vary substantially across studies. We hypothesize that part of this spread reflects protocol heterogeneity—differences in evaluation and synthesis choices such as leakage-safe data splitting, inverse-variance weighting, treatment stratification, dose–response specification, and missing-data policy. Using BiocharDS_V1.0, an open evidence synthesis table of biochar experiments, we define five representative protocols (P1–P5) and compare them while holding the model fixed (linear regression). We report cross-validated predictive error (RMSE), a "typical" effect defined as the model prediction at mean covariates (overall effect, OE), and stability diagnostics (fold-to-fold variability and sign flips of the rate coefficient).Protocol choices alone shift the typical SOC response from OE ≈ 0.18 to 0.28 lnRR (a drift of ≈ 0.10 lnRR), corresponding to roughly 20–32% SOC increase at mean conditions, despite identical data and model family. Leakage-prone row-wise splitting produces lower and more stable-looking error estimates, whereas study-grouped validation reveals substantially higher uncertainty, consistent with strong within-study dependence (under row-wise splitting, ~98% of test rows share a study identifier with training). Inverse-variance weighting substantially reduces the typical effect, whereas treatment-type stratification shifts the headline effect and its uncertainty by changing the target regime; added flexibility (quadratic dose–response) increases error and destabilizes rate signals. We provide a minimal reporting checklist and recommend protocol-driven sensitivity analysis (including missing-data sensitivity) as a lightweight standard for biochar SOC evidence synthesis and technology-relevant reporting.

Cover Letter

Date: 2026-02-22
To: The Editor-in-Chief
Journal: *Bioresource Technology*

Dear Editor,

We would like to submit our manuscript entitled "*Protocol choices contribute to variability in reported biochar impacts on soil organic carbon: a reproducible open-data study with BiocharDS*" for consideration as a Research Article in *Bioresource Technology*.

In this work, we use the open BiocharDS dataset and a controlled protocol comparison (holding the model fixed) to quantify how evaluation choices — such as leakage-prone versus study-grouped validation, inverse-variance weighting, treatment-type stratification, dose–response flexibility, and missing-data policy — can shift both the estimated "typical" SOC response to biochar and the apparent generalization performance.

We believe the manuscript is relevant to the BT audience because it addresses a practical source of uncertainty that directly affects technology-relevant claims drawn from evidence syntheses. The submission emphasizes transparency and reproducibility and provides a minimal protocol reporting checklist and supporting materials to reproduce the main tables and figures.

Statements

- This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere.
- All authors have read and approved the submitted version.
- The authors declare no competing interests.
- Data and code availability: see the Data availability statement in the manuscript and the reproducibility materials in this submission package.

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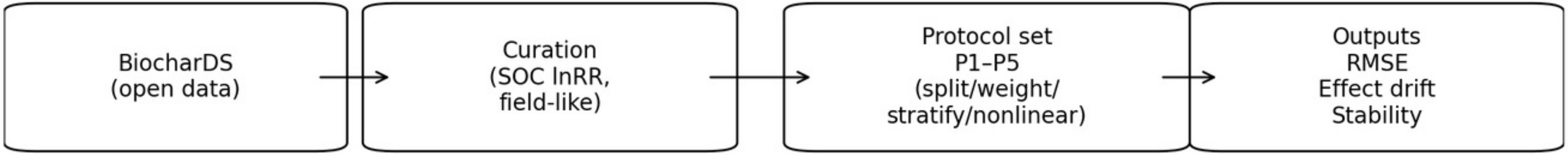
Thank you for considering our submission.

Sincerely,

Jincheng Li (on behalf of all authors)

Graphical Abstract (for review)



Highlights (for review)

• Protocol choices (split, weighting, stratification) shift biochar SOC conclusions.
• Grouped CV by study reveals optimistic bias under row-wise evaluation.
• Inverse-variance weighting reduces estimated SOC gains in BiocharDS.
• A minimal protocol checklist enables reproducible, leakage-safe evidence synthesis.

TITLE PAGE (non-anonymized)

Bioresource Technology

Title

Protocol choices contribute to variability in reported biochar impacts on soil organic carbon: a reproducible open-data study with BiocharDS

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data and code are provided in the Supplementary materials and will be made publicly available upon acceptance (e.g., via a repository with DOI).

Protocol choices contribute to variability in reported biochar impacts on soil organic carbon: a reproducible open-data study with BiocharDS

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ABSTRACT

Global syntheses of biochar impacts on soil organic carbon (SOC) are increasingly used to justify climate mitigation and soil management, yet reported effect sizes vary substantially across studies. We hypothesize that part of this spread reflects *protocol heterogeneity*—differences in evaluation and synthesis choices such as leakage-safe data splitting, inverse-variance weighting, treatment stratification, dose–response specification, and missing-data policy. Using BiocharDS_V1.0, an open evidence synthesis table of biochar experiments, we define five representative protocols (P1–P5) and compare them while holding the model fixed (linear regression). We report cross-validated predictive error (RMSE), a "typical" effect defined as the model prediction at mean covariates (overall effect, OE), and stability diagnostics (fold-to-fold variability and sign flips of the rate coefficient).

Protocol choices alone shift the typical SOC response from OE $\approx$ 0.18 to 0.28 lnRR (a drift of $\approx$ 0.10 lnRR), corresponding to roughly 20–32% SOC increase at mean conditions, despite identical data and model family. Leakage-prone row-wise splitting produces lower and more stable-looking error estimates, whereas study-grouped validation reveals substantially higher uncertainty, consistent with strong within-study dependence (under row-wise splitting, ~98% of test rows share a study identifier with training). Inverse-variance weighting substantially reduces the typical effect, whereas treatment-type stratification shifts the headline effect and its uncertainty by changing the target regime; added flexibility (quadratic dose–response) increases error and destabilizes rate signals. We provide a minimal reporting checklist and recommend protocol-driven sensitivity analysis (including missing-data sensitivity) as a lightweight standard for biochar SOC evidence synthesis and technology-relevant reporting.

1. Introduction

Biochar, a carbon-rich product of biomass pyrolysis and related thermochemical pathways, is increasingly considered in soil-amendment strategies for carbon sequestration (Lehmann and Joseph, 2015). Because biochar can persist in soils and interact with nutrient cycling, its net impact on soil organic carbon (SOC) is both scientifically and technologically relevant (Lehmann and Joseph, 2015). Accordingly, a large primary-experiment literature has been synthesized through global meta-analyses and open evidence datasets (Li et al., 2024; Xu et al., 2021; Chagas et al., 2022). Yet, even among recent syntheses, headline SOC benefits are not fully consistent: reported "typical" SOC responses differ by tens of percentage points, complicating inference and downstream carbon-accounting applications.

Such disagreement is often attributed to genuine environmental heterogeneity (soil type, climate, feedstock, pyrolysis conditions), but an additional driver is methodological: different studies implicitly adopt different *evaluation protocols*. In this paper, a "protocol" is the bundle of analytical choices that maps an open synthesis table to a headline effect and an associated uncertainty statement. In biochar SOC evidence synthesis, protocol components commonly include (i) the unit of analysis (row-level comparisons vs study-level generalization), (ii) the split rule used for evaluation, (iii) whether and how to apply inverse-variance weighting, (iv) whether biochar-only interventions are pooled with co-applied amendments, (v) dose–response flexibility, and (vi) missing-data policy. Because these components are inconsistently specified and rarely reported in a standardized way, it is difficult to diagnose why two syntheses disagree or to attribute differences to data versus protocol.

The problem is acute for open synthesis datasets that contain multiple treatment–control comparisons per study (Li et al., 2024). Treating rows as independent observations is generally incorrect, and naive row-wise train/test splitting can create information leakage: the same study can contribute rows to both training and test sets, allowing models to exploit study-specific signals (Roberts et al., 2017; Varma and Simon, 2006). In our curated BiocharDS cropping subset, this leakage is almost deterministic under row-wise K-fold splitting: on average, ~98% of test rows (and ~97% of test studies) share a study identifier with the corresponding training fold. This means that row-wise evaluation is poorly aligned with the common scientific goal of generalizing to new studies, and can understate uncertainty.

Weighting choices are similarly consequential. Inverse-variance weighting is standard in classical random-effects meta-analysis (DerSimonian and Laird, 1986; Hedges et al., 1999), but in open synthesis tables the implied weights can be extremely concentrated. Using the standard lnRR variance approximation, the top 5% of comparisons in our curated table account for ~66% of total inverse-variance weight, so weighting is not a minor refinement: it can pivot the estimand toward a small subset of high-precision comparisons.

Finally, missing-data handling is unavoidable in global syntheses. Even after restricting to cropping entries with SOC lnRR, key covariates remain partially observed (e.g., mean annual temperature, MAT, missing in ~39% of entries and mean annual precipitation, MAP, missing in ~28%). Different missingness policies (complete-case analysis, single imputation, multiple imputation) trade bias and variance differently, and should be treated as part of the protocol rather than an afterthought.

Here we introduce a protocol-driven evidence synthesis workflow that makes protocol assumptions explicit and measurable. Using BiocharDS_V1.0 (Li et al., 2024), we instantiate five representative protocols (P1–P5) that vary leakage-proneness, weighting, treatment stratification, and dose–response flexibility while holding the model fixed. We quantify protocol-induced drift in a typical SOC effect estimate, along with protocol-induced changes in out-of-sample error and stability. We also provide a minimal reporting checklist and reproducible scripts to support future biochar SOC assessments and to make disagreements across studies scientifically interpretable.

2. Materials and methods

2.1. Open dataset and study unit

We used the BiocharDS_V1.0 open dataset (Li et al., 2024) and focused on SOC effects reported as log response ratios (lnRR). Each row in the curated analysis table represents one treatment–control comparison extracted from an original study, and multiple rows may originate from the same study. This clustered structure motivates treating the *study* as the unit of generalization when evaluating models.

2.2. Outcome, covariates, and variance proxy

The primary outcome was SOC lnRR, denoted as y (Hedges et al., 1999). For a treatment mean $\bar{Y}_T$ and control mean $\bar{Y}_C$, we use the standard definition

$$\mathrm{lnRR} = \log\left(\frac{\bar{Y}_T}{\bar{Y}_C}\right). \tag{1}$$

When raw means, standard deviations (s_T, s_C) and sample sizes (n_T, n_C) were available, we approximated the sampling variance of lnRR via a common delta-method form

$$\mathrm{Var}(\mathrm{lnRR}) \approx \frac{s_T^2}{n_T\,\bar{Y}_T^2} + \frac{s_C^2}{n_C\,\bar{Y}_C^2}, \tag{2}$$

and used it for inverse-variance weighting, a standard choice in meta-analysis (DerSimonian and Laird, 1986).

Core covariates were selected a priori to represent biochar input and site context: biochar application rate (t ha^{-1}), biochar pyrolysis temperature (°C), baseline soil pH, and climate proxies (mean annual temperature, MAT; and mean annual precipitation, MAP).

2.3. Curation rules and missing-data handling

We restricted the main analysis to entries with non-missing SOC lnRR and applied a cropping proxy (entries with a non-empty crop type were labeled as cropping). Treatment codes were collapsed into two categories: B for biochar-only and BF for biochar with co-applied fertilizer or other amendments, following the dataset coding (Li et al., 2024).

Covariate missingness is substantial in global synthesis tables. Prior to imputation in our curated table, missing rates were ~17% for pyrolysis temperature, ~14% for baseline soil pH, ~39% for mean annual temperature (MAT), and ~28% for mean annual precipitation (MAP); by construction, biochar application rate was fully observed (required for inclusion).

Table 1
Descriptive statistics of the curated analysis table (cropping subset).

variable	count	mean	std	min	25%	50%	75%	max	missing_rate
rate	796.000	21.297	23.885	0.150	5.000	13.500	25.000	225.000	0.000
pyrolysis_temp	796.000	458.310	76.280	230.000	450.000	450.000	500.000	900.000	0.166
soil_pH0	796.000	6.598	1.285	3.840	5.860	6.300	8.000	9.340	0.141
MAT	796.000	15.361	4.416	4.500	14.100	15.400	16.000	32.000	0.389
MAP	796.000	928.740	456.623	45.000	630.000	852.000	1177.000	2618.000	0.279
y	796.000	0.278	0.336	-1.570	0.060	0.190	0.420	1.620	0.000

Notes: rate is biochar application rate (t ha^{-1}); pyrolysis_temp is biochar pyrolysis temperature (°C); soil_pH0 is baseline soil pH; MAT is mean annual temperature (°C); MAP is mean annual precipitation (mm); y is SOC lnRR. Summary statistics are computed after median imputation; `missing_rate` is the fraction missing before imputation. Pre-imputation missing rates were: pyrolysis_temp 0.166, soil_pH0 0.141, MAT 0.389, MAP 0.279 (rate and y complete).

For the primary analysis, we used an indicator strategy: for each covariate, we created a binary missing flag and imputed missing values with the column median. Missingness indicators were included as covariates in the model so that fitted response surfaces can partially adapt to systematic missingness patterns without discarding rows.

Because missing-data handling is itself a protocol component that can change both the sample and the estimand, we performed a sensitivity analysis under a complete-case policy (dropping any row with missing core covariates; $N = 335$) and report the resulting protocol summaries in the Supplementary Information (Table S1).

This study is a secondary re-analysis of an open dataset rather than a de novo systematic review; therefore PRISMA (Page et al., 2021) items related to database searching and screening flow are not directly applicable. To strengthen transparency and facilitate auditing, we provide a PRISMA-applicability checklist (Table S3), a curation report, and a field-level data dictionary in the Supplementary Information.

The main curated table contained $N = 796$ entries from $S = 140$ studies. Descriptive statistics of the curated analysis table are provided in Table 1.

2.4. Protocol set (what we compare)

Our goal was to compare *protocols* rather than algorithms. All protocols used the same base model, and only protocol components were changed. Each protocol is defined by a bundle of four choices: (i) split rule (row-wise vs grouped by study), (ii) inverse-variance weighting (DerSimonian and Laird, 1986), (iii) treatment-type stratification (B vs BF), and (iv) dose–response flexibility (linear vs quadratic in application rate). Table 2 summarizes the five representative protocols.

To make the comparison concrete for readers, the five protocols can be interpreted as a progression from a naive row-wise evaluation (P1) toward increasingly leakage-safe and "meta-analysis-like" variants (P2–P4), with an additional flexibility stress-test (P5):

- **P1**: row-wise K-fold split; unweighted; pooled B and BF; linear main effects.
- **P2**: grouped split by `study_id` (leakage-safe); unweighted; pooled; linear main effects.
- **P3**: P2 + inverse-variance weighting (weighted least squares).
- **P4**: P3 + treatment-type stratification (effects reported for the B regime).
- **P5**: P4 + quadratic dose–response in application rate (adds $\mathtt{rate}^2$).

2.5. Model specification (fixed across protocols)

To isolate protocol effects, we fixed the model to a transparent linear specification and changed *only* protocol components. Let x_i denote the covariate vector for entry i (rate, pyrolysis temperature, baseline soil pH, MAT, MAP, and missingness indicators). For protocols P1–P4 we fit

$$y_i = \beta_0 + x_i^\top \beta + \varepsilon_i, \tag{3}$$

Table 2
Protocol set compared in this study (P1–P5).

Protocol	Split	Weight	B vs BF	Dose	Key idea
P1	Row-wise	none	No	linear	Naive row-wise splitting (leakage-prone).
P2	Study-grouped	none	No	linear	Leakage-safe evaluation by holding out entire studies (`GroupKFold` by `study_id`).
P3	Study-grouped	$1/v$	No	linear	P2 + inverse-variance weighting (classic meta weighting).
P4	Study-grouped	$1/v$	Yes	linear	P3 + treatment-type stratification; headline effect reported for biochar-only (B; BF=0).
P5	Study-grouped	$1/v$	Yes	quadratic	P4 + quadratic dose–response in application rate (adds rate^2).

Notes: v denotes the effect-size variance proxy (Eq. 2), and inverse-variance weights are $w_i = 1/v_i$. All protocols use the same linear regression model; differences reflect protocol choices (split rule, weighting, stratification, and dose–response specification) rather than algorithm choice.

using ordinary least squares (unweighted protocols) or weighted least squares (inverse-variance protocols). For P5 we add a quadratic term in application rate, rate^2, as a simple flexibility stress-test.

When treatment-type stratification is enabled (P4–P5), we fit a fully interacted linear model with a treatment indicator (BF) so that biochar-only (B) and co-applied (BF) entries can follow different response surfaces (equivalent to separate fits). We report the standardized B-regime headline effect by setting BF=0.

2.6. Evaluation metrics, drift, and stability diagnostics

For each protocol, we performed 5-fold cross-validation and reported test-set root mean squared error (RMSE). For grouped protocols, we used GroupKFold with `study_id` as the grouping key to avoid leakage of study-specific signals across folds (Roberts et al., 2017; Varma and Simon, 2006). We quantify leakage under row-wise splitting as the fraction of test rows whose `study_id` appears in the training fold (averaged across folds), and we also report the analogous fraction of test studies. We quantify weight concentration under inverse-variance weighting as the share of total inverse-variance weight carried by the top 5% of rows. These audit summaries are reported in Table S4. RMSE is defined as

$$\mathrm{RMSE} = \sqrt{\frac{1}{n_{\mathrm{test}}} \sum_{i \in \mathrm{test}} \left(y_i - \hat{y}_i\right)^2}. \tag{4}$$

For stratified protocols (P4–P5), predictions $\hat{y}_i$ are generated using the appropriate stratum-specific fit for each test row, and RMSE is computed over all test rows; by contrast, OE is standardized to the biochar-only (B) regime (BF$=0$) for comparability of the headline effect.

In each fold, we computed an interpretable "overall effect" as the model prediction at the mean covariate vector $\bar{x}$ (and BF$=0$ when stratification is enabled):

$$\mathrm{OE} = \hat{y}(\bar{x}). \tag{5}$$

For interpretability, we report the implied percent SOC change at mean conditions as $100\,(\exp(\mathrm{OE}) - 1)$.

To make protocol sensitivity explicit, we define protocol-induced drift between two protocols a and b as

$$\Delta_{a \to b} = \mathrm{OE}_b - \mathrm{OE}_a. \tag{6}$$

We summarize drift through pairwise differences of OE means (reported in the SI) and interpret these differences relative to typical effect magnitudes in the biochar SOC literature.

To quantify stability, we report fold-to-fold standard deviations of RMSE and OE, and we compute a sign-flip rate for the estimated *linear* rate coefficient across folds, defined as the fraction of adjacent folds in which the estimated sign changes:

$$\mathrm{Flip} = \frac{1}{K-1} \sum_{k=1}^{K-1} \mathbb{I}\left[\mathrm{sign}(\hat{\beta}_{\mathrm{rate}}^{(k)}) \neq \mathrm{sign}(\hat{\beta}_{\mathrm{rate}}^{(k+1)})\right], \tag{7}$$

where K is the number of folds. This diagnostic highlights when dose–response signals are unstable under protocol changes or limited information.

2.7. Reproducibility and implementation

All protocols share the same curated table and the same feature preprocessing. Protocol definitions are stored as explicit configuration objects (split rule, weighting flag, stratification flag, and dose-response flag), and the full pipeline outputs both (i) fold-level diagnostics and (ii) aggregated summary tables and figures. To facilitate independent checking, we provide one-click scripts in an executable reproducibility snapshot (Research Data external item; `https://github.com/kqxsxlj/bt-reproduce`) that rebuild Tables 1–3 and Figs. 1–3 from the curated table.

3. Results and discussion

3.1. Curated dataset overview

The main curated table contained $N = 796$ entries from $S = 140$ studies, including 579 biochar-only (B) and 217 co-applied (BF) entries. Table 1 summarizes the outcome and key covariates after curation and imputation. The SOC lnRR outcome had mean 0.278 and median 0.208, corresponding to an average SOC increase of about 32% at the scale of exp(lnRR).

Biochar application rates were right-skewed (median 13.5 t ha^{-1}; max 225 t ha^{-1}), and pyrolysis temperatures centered around 450°C. The outcome distribution includes both positive and negative lnRR values, reflecting substantial heterogeneity across sites and designs.

Within-study clustering is substantial: most studies contribute only a few entries, but a small number contribute many comparisons, leading to a long-tailed distribution of entries per study. This structure is visualized in Fig. 1 and motivates grouped evaluation by study to avoid leakage.

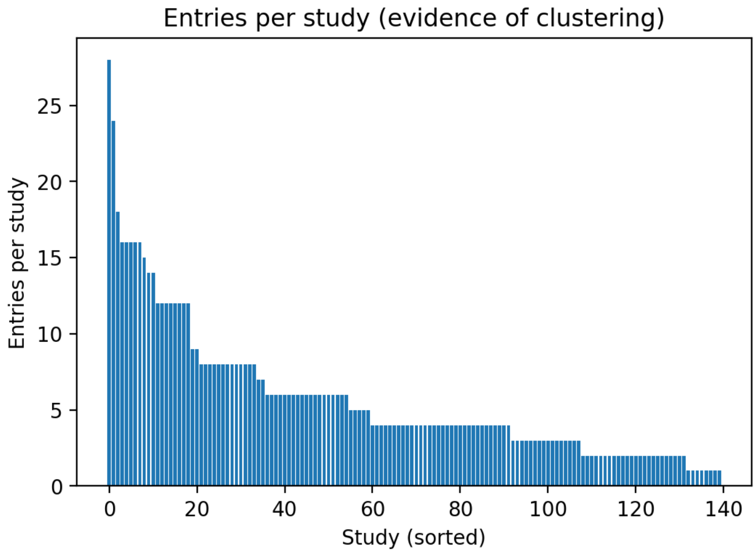

Figure 1: Within-study clustering in BiocharDS (cropping subset). Bars show the number of treatment–control comparisons contributed by each study ($N = 796$ entries across $S = 140$ studies), sorted by study. The heavy-tailed pattern (few studies contribute many rows) illustrates why row-wise random splits can leak study-specific signals into both train and test sets, motivating grouped cross-validation at the study level.

3.2. Protocol comparison across five protocols

Table 3 summarizes cross-validated RMSE and protocol-dependent summaries of a typical SOC effect (overall effect at mean covariates). To aid interpretation, Table 3 also reports the implied percent SOC change at mean conditions. Below we highlight three main patterns: (i) leakage-safe grouping changes both the *level* and *variability* of error estimates; (ii) weighting and stratification shift the typical effect and its implied estimand; and (iii) added flexibility can destabilize dose–response signals.

Table 3
Protocol summary (5-fold cross-validation on cropping subset).

protocol	rmse_mean	rmse_sd	overall_mean	overall_sd	rate_sign_flip	pct
P1	0.287	0.038	0.278	0.004	0.000	32.010
P2	0.318	0.104	0.275	0.016	0.000	31.643
P3	0.348	0.124	0.180	0.017	0.000	19.689
P4	0.351	0.116	0.189	0.021	0.000	20.832
P5	0.392	0.113	0.198	0.027	0.500	21.904

Notes: rmse_mean and rmse_sd are the mean and SD of test RMSE across folds. For stratified protocols (P4–P5), RMSE is computed over all test rows using the appropriate stratum-specific fit for each row. overall_mean and overall_sd are the mean and SD of the "overall effect" (model prediction at mean covariates) across folds; when treatment-type stratification is enabled (P4–P5), the headline effect is reported for biochar-only (B; BF=0). rate_sign_flip is the fraction of adjacent fold transitions in which the sign of the estimated linear rate coefficient changes. pct is $100\,(\exp(\text{overall\_mean}) - 1)$.

Leakage-safe evaluation reveals higher uncertainty. Row-wise evaluation (P1) produced a lower and more stable-looking error estimate (RMSE $= 0.287 \pm 0.038$), whereas leakage-safe grouped evaluation by study (P2) increased both the mean and variability of error (RMSE $= 0.318 \pm 0.104$). This is consistent with substantial study-level dependence: under a naive row-wise split, ~98% of test rows share a study identifier with the corresponding training fold, so row-wise evaluation can partially "memorize" study-specific signals and understate uncertainty. Grouped validation therefore better aligns with the scientific goal of generalizing to *new studies*.

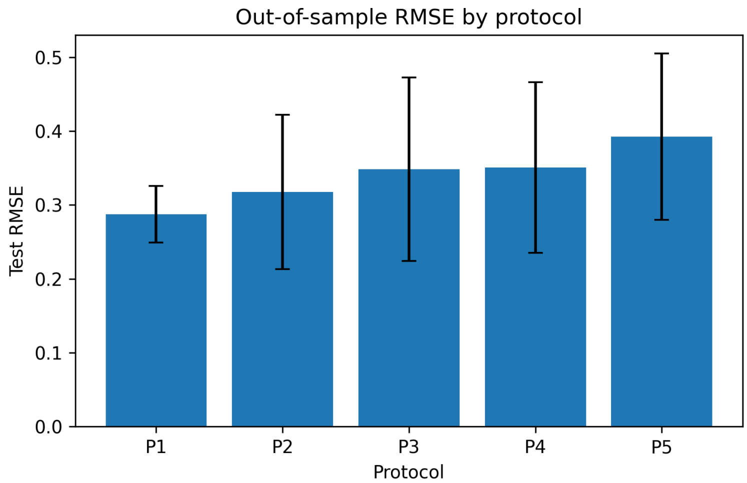

Figure 2: Protocol sensitivity of predictive error. Cross-validated test RMSE for protocols P1–P5 on the cropping subset. Bars show the mean RMSE across 5 folds and error bars show ± 1 SD across folds. Protocols differ only in evaluation components (split rule, weighting, stratification, and dose–response specification; see Table 2), so differences reflect protocol choice rather than algorithm choice.

Weighting and stratification change the typical effect. When inverse-variance weighting was enabled (P3), mean RMSE increased to 0.348 and the typical SOC effect estimate dropped from lnRR≈0.275 to 0.180 (about 32% to 20% SOC increase). Adding treatment-type stratification (P4) produced a similar RMSE (0.351) and a slightly higher typical effect (lnRR≈0.189).

A practical interpretation is that weighting and stratification are not merely "statistical refinements": they alter which entries dominate the fit and which regime is used for the headline effect. In our curated table, inverse-variance weights are highly concentrated (the top 5% of comparisons contribute ~66% of total weight), so weighted protocols can pivot the estimand toward a small set of high-precision comparisons. Combined with the strong clustering visible in Fig. 1, these choices can substantially shift the implied typical response.

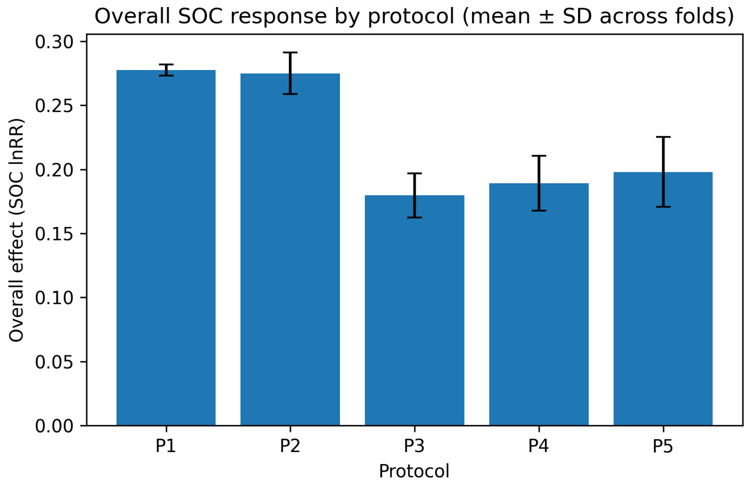

Figure 3: Protocol sensitivity of the typical SOC effect. Typical SOC response to biochar across protocols, summarized as the "overall effect" (OE), i.e., the model prediction at the mean covariate vector (BF fixed to 0 when stratification is enabled). Bars show the mean OE (in lnRR) across folds and error bars show ±1 SD. Percent SOC change at mean conditions equals $100\,(\exp(\mathrm{OE}) - 1)$ (reported in Table 3).

Flexibility can destabilize dose–response signals. Allowing a quadratic dose–response in application rate (P5) further increased RMSE (0.392) and led to an unstable linear rate effect: the sign of the rate coefficient flipped across adjacent folds in 50% of fold transitions. This suggests that when protocols introduce additional flexibility, stability diagnostics (such as coefficient sign consistency) should be reported alongside point estimates.

Practical interpretation of protocol-induced drift. Across protocols P1–P5, the implied SOC increase at mean conditions ranges from about 20% (P3) to about 32% (P1). For many application contexts, this spread is large enough to affect both scientific interpretation (e.g., "moderate" vs "strong" benefits) and downstream techno-economic or climate-relevance assessments. Importantly, this drift arises without changing the underlying data or model family—only the protocol changes.

3.3. Implications for biochar SOC evidence synthesis

Together, these results show that protocol decisions can materially change conclusions even before model choice. In practice, authors often vary multiple protocol components implicitly, which makes it difficult to reconcile why different studies or meta-analyses report different "overall" biochar SOC benefits. Our protocol-driven framing turns this into a measurable sensitivity problem: *how much does the headline effect move when only protocol assumptions change?*

Reconciling conflicting meta-analytic estimates. A key practical consequence is that protocol heterogeneity can explain a substantial share of the spread in published SOC estimates. For example, a synthesis that effectively pools all comparisons and evaluates row-wise (analogous to P1/P2) can yield a headline increase near 32% at mean conditions, whereas a synthesis that emphasizes classical meta-analytic weighting and separates biochar-only from co-applied amendments (analogous to P3/P4) yields a headline increase near 20–21%. This ~12 percentage-point gap arises on the *same* curated table with the *same* model, indicating that at least part of the disagreement across published syntheses is plausibly methodological rather than purely ecological.

Implications for carbon accounting and carbon markets. The same drift matters in applied settings because SOC benefits are often converted to carbon-removal claims and then to credits or policy targets. Moving from a 20% to a 32% SOC increase corresponds to a 1.6× difference in the implied SOC gain (~60% relative). Under accounting frameworks where credited removals scale approximately linearly with estimated SOC change (holding other accounting factors constant), protocol choice can therefore translate into large swings in credited CO_2-equivalent removals for the same biochar deployment. This does not imply that one protocol is "correct" in all contexts; rather, it motivates reporting protocol sensitivity alongside point estimates whenever results are used for techno-economic assessment, carbon-credit issuance, or climate-policy arguments.

Methodological robustness: missingness and clustering. Two additional sources of protocol sensitivity deserve explicit attention. First, missing-data policy is unavoidable in global syntheses. In a complete-case sensitivity analysis that drops any row with missing core covariates ($N = 335$), headline SOC effects shift downward (unweighted protocols yield ~25% and weighted protocols ~17% SOC increase), but the qualitative pattern of protocol-induced drift remains (Table S1). Second, the dataset is strongly clustered by study. Our main comparison enforces study-level generalization through GroupKFold, which addresses the most direct form of information leakage for predictive evaluation (Roberts et al., 2017; Varma and Simon, 2006). A full multilevel mixed-effects meta-analytic model is a valuable complementary approach for estimation and inference; as an exploratory check, a random-intercept mixed-effects model fitted to the same curated table yields an overall biochar-only effect at mean conditions that is close to the grouped unweighted estimate (OE ~0.27 lnRR; Table S2), suggesting that our grouped-evaluation framing captures a major part of the study-level dependence for the purpose of protocol comparison.

Finally, although PRISMA (Page et al., 2021) is the gold standard for reporting de novo systematic reviews and meta-analyses, our work is a secondary analysis of an existing open dataset rather than a new literature search. We therefore report a PRISMA-applicability checklist (Table S3) and transparent curation documentation in the Supplementary Information, and we emphasize that our primary contribution is *protocol sensitivity*, not a new definitive global SOC estimate.

Minimum protocol reporting checklist. When making technology-relevant claims about biochar SOC benefits from open synthesis datasets, we recommend reporting at least:

- **Split rule and unit of generalization:** row-wise vs study-grouped; how clusters are identified.
- **Weighting:** whether inverse-variance weighting is used; how effect-size variances are approximated; whether weights are stabilized or normalized.
- **Treatment handling:** whether biochar-only and co-applied treatments are pooled or stratified; which regime is used for the headline effect.
- **Dose–response specification:** linear vs non-linear terms; justification for flexibility; and stability diagnostics for inferred rate effects.
- **Missing-data policy:** complete-case vs imputation strategy; and whether missingness indicators are used.
- **Uncertainty reporting:** variability of RMSE and typical effect across folds (not only mean values).

Sensitivity analysis as standard practice. Rather than selecting a single "best" protocol, we recommend stress-testing conclusions across a small set of representative protocols (such as P1–P5) and, when feasible, across plausible missing-data policies. This protocol-driven sensitivity analysis is lightweight, easy to reproduce, and directly targets a common source of disagreement across studies: hidden analytical choices.

A pragmatic default for headline reporting. While the "right" protocol depends on the scientific question, our results suggest a practical baseline for technology-relevant reporting from clustered open datasets: (i) use study-grouped splits for any generalization claim, (ii) report both unweighted and inverse-variance weighted estimates, (iii) avoid pooling materially different treatment regimes without either stratification or explicit interactions, and (iv) report missing-data sensitivity when covariate missingness is substantial.

Suggested reporting template. To make results comparable across studies, a short protocol statement can be included in the Methods or SI, for example: "We evaluate on BiocharDS using 5-fold study-grouped cross-validation (group=study_id). We fit a linear model with (rate, pyrolysis temperature, soil pH, MAT, MAP) and missingness indicators, and we report both unweighted and inverse-variance weighted fits. The headline effect is the predicted lnRR at mean covariates for biochar-only (BF=0), with uncertainty summarized as mean±SD across folds."

Scope and limitations. Our protocol set is not exhaustive and we fixed the model (linear regression) to isolate protocol effects. Nonetheless, the observed drift suggests that protocol uncertainty should be reported alongside model uncertainty. Future work can extend this framework to additional outcomes, non-linear learners, and fully hierarchical meta-analytic models (e.g., multilevel random-effects and robust-variance approaches) while retaining protocol sensitivity as the organizing principle.

4. Conclusions

Using BiocharDS_V1.0 and a controlled comparison in which the model is fixed, we show that evaluation-protocol choices alone can shift the estimated typical SOC response to biochar from OE ≈ 0.18 to 0.28 lnRR (a drift of ≈ 0.10 lnRR), corresponding to ~20–32% SOC increase at mean conditions. Leakage-prone row-wise splits produce overly stable and optimistic performance estimates, whereas study-grouped validation reveals substantially higher uncertainty, highlighting the need to treat study as the unit of generalization. Inverse-variance weighting substantially reduces the typical effect, and treatment-type stratification shifts the headline effect and its uncertainty by changing the target regime (biochar-only vs co-applied), indicating that these common choices alter the target estimand rather than merely improving precision. Allowing more flexible dose–response forms can further increase prediction error and yield unstable rate effects, suggesting that flexibility should be justified and accompanied by stability diagnostics. Finally, because covariate missingness is substantial in global syntheses, missing-data policy should be treated as a protocol component and reported transparently (with sensitivity analysis where feasible).

We therefore recommend that future technology-relevant claims about biochar SOC benefits report, at minimum, the split rule, clustering/unit of generalization, weighting scheme, treatment stratification, dose–response specification, and missing-data policy, and that conclusions be stress-tested across a small, representative protocol set.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

BiocharDS_V1.0 is openly available at DOI: 10.1038/s41597-023-02867-9. An executable reproducibility snapshot (scripts, protocol configurations, and processed analysis tables to regenerate Tables 1–3 and Figs. 1–3) is available at `https://github.com/kqxsxlj/bt-reproduce` (snapshot ZIP: `riec_bt_reproduce.zip`).

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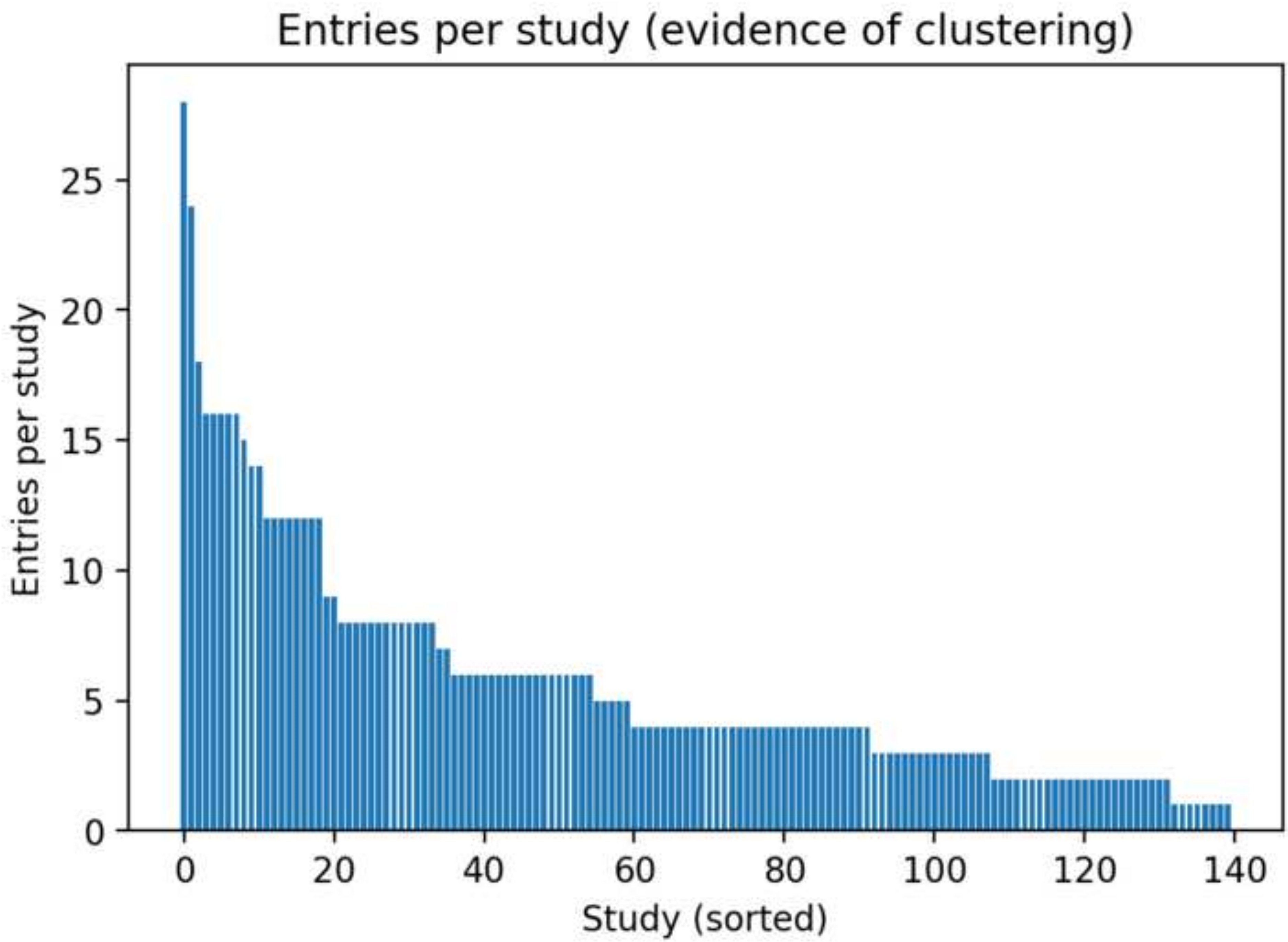
variable	count	mean	std	min	25%	50%	75%
rate	796.000	21.297	23.885	0.150	5.000	13.500	25.000
pyrolysis_temp	796.000	458.310	76.280	230.000	450.000	450.000	500.000
soil_pH0	796.000	6.598	1.285	3.840	5.860	6.300	8.000
MAT	796.000	15.361	4.416	4.500	14.100	15.400	16.000
MAP	796.000	928.740	456.623	45.000	630.000	852.000	1177.000
y	796.000	0.278	0.336	-1.570	0.060	0.190	0.420

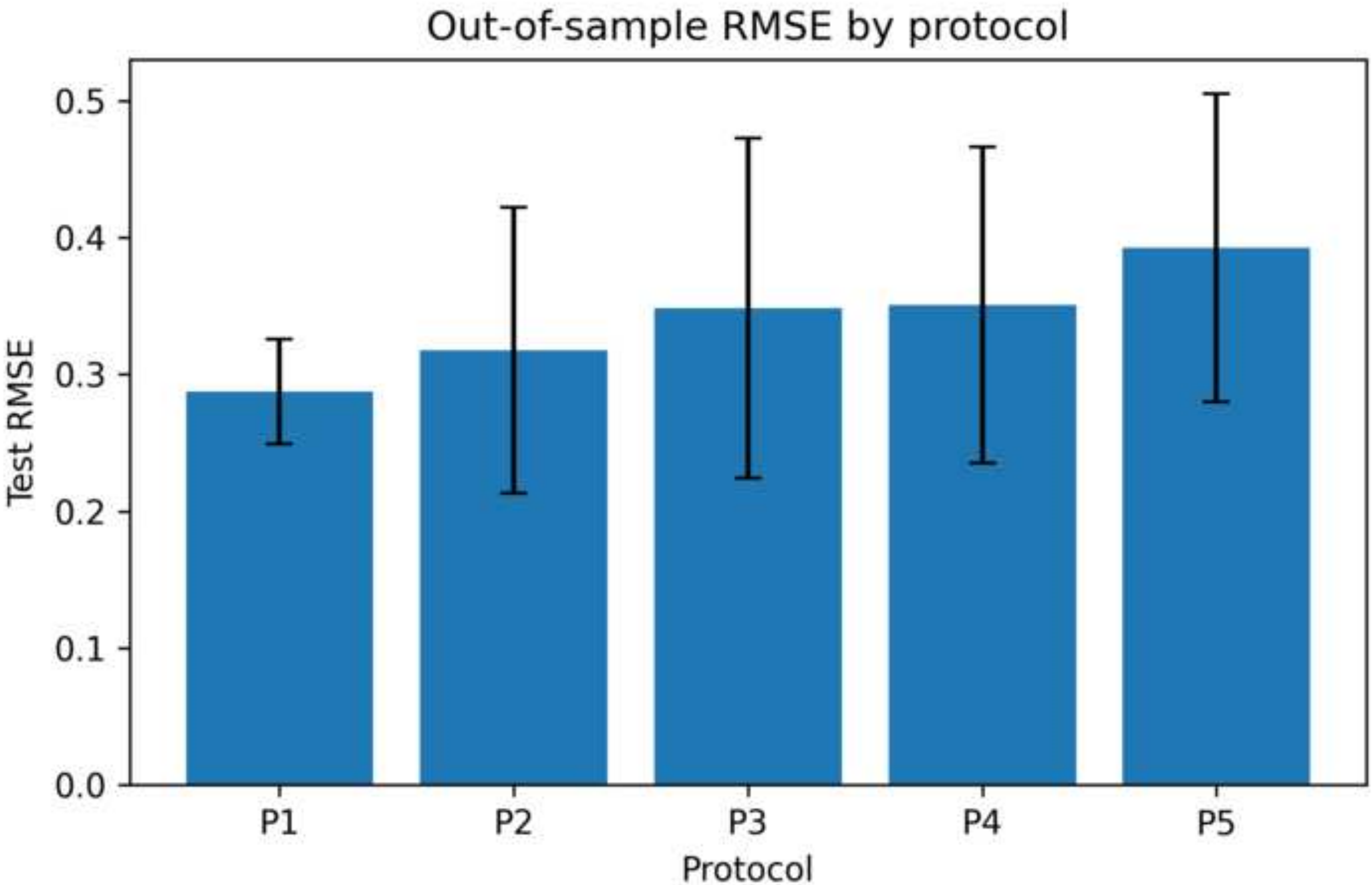
max	missing_rate
225.000	0.000
900.000	0.166
9.340	0.141
32.000	0.389
2618.000	0.279
1.620	0.000

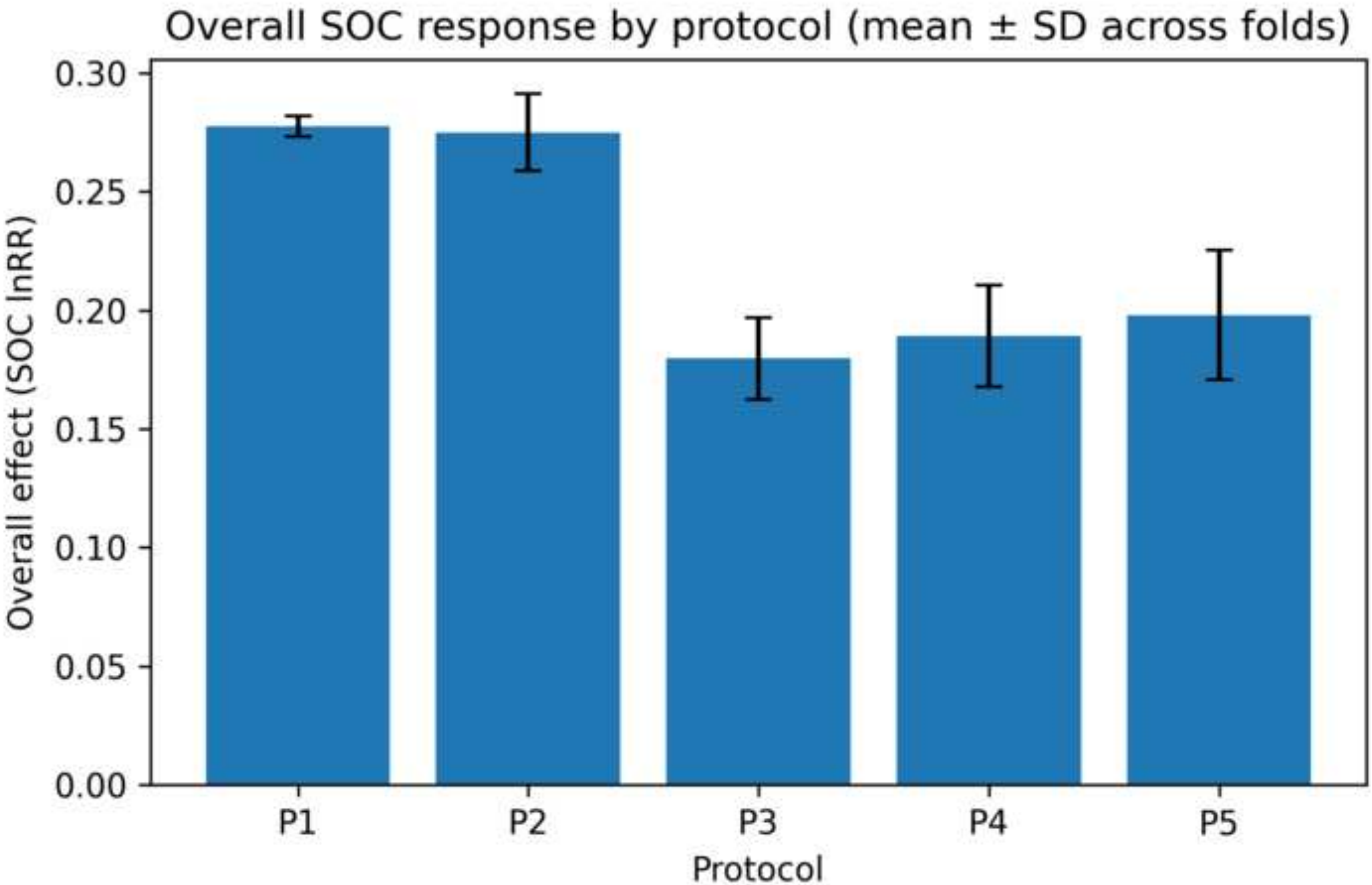
Protocol	Split	Weight	B vs BF	Dose
P1	Row-wise	none	No	linear
P2	Study-grouped	none	No	linear
P3	Study-grouped	1/v	No	linear
P4	Study-grouped	1/v	Yes	linear
P5	Study-grouped	1/v	Yes	quadratic

Key idea
Naive row-wise splitting (leakage-prone).
Leakage-safe evaluation by holding out entire studies (\texttt{GroupKFold} by \texttt{study_id}).
P2 + inverse-variance weighting (classic meta weighting).
P3 + treatment-type stratification; headline effect reported for biochar-only (B; BF=0).
P4 + quadratic dose--response in application rate (adds \mathrm{rate}^2).

protocol	rmse_mean	rmse_sd	overall_mean	overall_sd	rate_sign_flip	pct
P1	0.287	0.038	0.278	0.004	0.000	32.010
P2	0.318	0.104	0.275	0.016	0.000	31.643
P3	0.348	0.124	0.180	0.017	0.000	19.689
P4	0.351	0.116	0.189	0.021	0.000	20.832
P5	0.392	0.113	0.198	0.027	0.500	21.904


Entries per study (evidence of clustering)
Entries per study
0
5
10
15
20
25
0
20
40
60
80
100
120
140
Study (sorted)


Out-of-sample RMSE by protocol
Test RMSE
0.0
0.1
0.2
0.3
0.4
0.5
P1
P2
P3
P4
P5
Protocol


Overall SOC response by protocol (mean ± SD across folds)
Overall effect (SOC lnRR)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
P1
P2
P3
P4
P5
Protocol

Declaration of Interest Statement

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT Author Statement

CRediT authorship contribution statement

Jincheng Li: Conceptualization; Methodology; Software; Formal analysis; Data curation; Validation; Visualization; Writing – original draft; Writing – review & editing.

Meiying Liu: Supervision; Writing – review & editing.

Doudou Jin: Data curation; Validation.

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